

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286476

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

HOSOYA; Mamoru

POWER GENERATION DEVICE

Abstract

Provided is a power generation device capable of facilitating electrical connection of a plurality of first electrode layers and electrical connection of a plurality of second electrode layers. A first chargeable layer **13** and a second chargeable layer **23** face each other due to parts of a plurality of first sheet members **10** and parts of a plurality of second sheet members **20** being alternately stacked, a power generation device **1** generates power by way of a change in a mutual contact state between the first chargeable layer **13** and the second chargeable layer **23**, a first connecting conductor **30** penetrates through another part of each of the plurality of first sheet members **10** and is connected to a first electrode layer **12** of each of the plurality of first sheet members **10**, and a second connecting conductor **40** penetrates through another part of each of the plurality of second sheet members **20** and is connected to a second electrode layer **22** of each of the plurality of second sheet members **20**.

Inventors: HOSOYA; Mamoru (Kanagawa, JP)

Applicant: TOKYO OHKA KOGYO CO., LTD. (Kanagawa, JP)

Family ID: 96949953

Appl. No.: 19/072264

Filed: March 06, 2025

Foreign Application Priority Data

JP	2024-037195	Mar. 11, 2024
JP	2025-025060	Feb. 19, 2025

Publication Classification

Int. Cl.: H02N1/04 (20060101)

U.S. Cl.:

Background/Summary

[0001] This application is based on and claims the benefit of priority from Japanese Patent Application Nos. 2024-037195 and 2025-025060, respectively filed on 11 Mar. 2024 and 19 Feb. 2025, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a power generation device that generates power using frictional charging.

Related Art

[0003] As a power generation device in the related art, a power generation device including a first member that includes a first chargeable layer on a side of an outer surface of a first electrode layer and a second member that includes a second chargeable layer on a side of an outer surface of a second electrode layer is known (see Japanese Unexamined Patent Application, Publication No. 2016-500248, for example). The power generation device in the related art has a structure in which the first member and the second member are stacked such that the first chargeable layer and the second chargeable layer face each other, and generates power by changing a contact state between the first chargeable layer and the second chargeable layer.

[0004] In order to increase a power generation output of the power generation device, it has been studied to arrange, in a stacking direction, a plurality of sets each composed of a first chargeable layer and a second chargeable layer facing each other by alternately stacking a plurality of first members and a plurality of second members. [0005] Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2016-500248

SUMMARY OF THE INVENTION

[0006] In the power generation device including the plurality of first members and the plurality of second members, the first electrode layers of the plurality of first members are connected in parallel or in series by a connecting wire, and the second electrode layers of the plurality of second members are connected in parallel or in series by a connecting wire. In this case, a state in which the connection wires are connected to the first electrode layers and the second electrode layers in advance is achieved, other members are stacked on the first electrode layers and the second electrode layers with the connecting wires connected thereto to thereby form the first members and the second members. Therefore, there is a concern that for the power generation device including the plurality of first members and the plurality of second members, an operation of connecting the connecting wires to the first electrode layers and the second electrode layers may become complicated, which may lead to an increase in manufacturing cost.

[0007] An object of the present invention is to provide a power generation device capable of facilitating electrical connection of a plurality of first electrode layers and electrical connection of a plurality of second electrode layers.

[0008] A power generation device according to the present invention includes: a plurality of first sheet members; a plurality of second sheet members; a first connecting conductor; and a second connecting conductor. Each of the first sheet members includes a first electrode layer and a first chargeable layer disposed on an outer surface of the first electrode layer, each of the second sheet members includes a second electrode layer and a second chargeable layer disposed on an outer surface of the second electrode layer, the first chargeable layer and the second chargeable layer face each other due to parts of the plurality of first sheet members and parts of the plurality of second sheet members being alternately stacked, the power generation device being configured to generate

power by way of a change in a mutual contact state between the first chargeable layer and the second chargeable layer, the first connecting conductor penetrates through another part of each of the plurality of first sheet members and is connected to the first electrode layer of each of the plurality of first sheet members, and the second connecting conductor penetrates through another part of each of the plurality of second sheet members and is connected to the second electrode layer of each of the plurality of second sheet members.

[0009] The power generation device according to the present invention preferably includes: a first insulating member surrounding an outer peripheral portion of the first connecting conductor, the outer peripheral portion being located between the plurality of first sheet members; and a second insulating member surrounding an outer peripheral portion of the second connecting conductor, the outer peripheral portion being located between the plurality of second sheet members.

[0010] In the power generation device according to the present invention, it is preferable that a ratio $Ti1/Ts2$ between a thickness $Ti1$ of the first insulating member and a thickness $Ts2$ of the second sheet members and a ratio $Ti2/Ts1$ between a thickness $Ti2$ of the second insulating member and a thickness $Ts1$ of the first sheet members are each 0.8 or more and 1.2 or less.

[0011] In the power generation device according to the present invention, one of the first chargeable layer and the second chargeable layer is preferably made of polyimide with a porous structure while the other is preferably made of polyamide.

[0012] In the power generation device according to the present invention, the first connecting conductor preferably includes a pair of penetrating portions penetrating through other parts of the plurality of first sheet members in a stacking direction, and a coupling portion extending in a plane direction of the first sheet members and coupling proximal end portions of the pair of penetrating portions to each other, and the second connecting conductor preferably includes a pair of penetrating portions penetrating through other parts of the plurality of second sheet members in a stacking direction, and a coupling portion extending in a plane direction of the second sheet members and coupling proximal end portions of the pair of penetrating portions to each other.

[0013] The power generation device according to the present invention preferably includes: a first conducting wire connected to the first connecting conductor; and a second conducting wire connected to the second connecting conductor, the first conducting wire is preferably sandwiched between the coupling portion of the first connecting conductor and the first sheet member, and the second conducting wire is preferably sandwiched between the coupling portion of the second connecting conductor and the second sheet member.

[0014] In the power generation device according to the present invention, the first connecting conductor is preferably a conductive thread-shaped member with which the plurality of first sheet members are sewn together, and the second connecting conductor is preferably a conductive thread-shaped member with which the plurality of second sheet members are sewn together.

[0015] The power generation device according to the present invention preferably includes: a first conducting wire connected to the first connecting conductor; and a second conducting wire connected to the second connecting conductor, the first conducting wire is preferably sandwiched between a thread-shaped member configuring the first connecting conductor and the first sheet members, and the second conducting wire is preferably sandwiched between a thread-shaped member configuring the second connecting conductor and the second sheet members.

[0016] The power generation device according to the present invention preferably includes: spacing retention members that retain spacings between the first and second sheet members facing each other in a stacking direction. Preferably, the spacing retention members are each disposed between an end portion of the first sheet member adjacent to the second insulating member and the second sheet member facing at least one side of the end portion in the stacking direction and/or between an end portion of the second sheet member adjacent to the first insulating member and the first sheet member facing at least one side of the end portion in the stacking direction.

[0017] According to the present invention, it is possible to connect a plurality of first electrode

layers to a first connecting conductor and to connect a plurality of second electrode layers to a second connecting conductor by causing the first connecting conductor to penetrate through a plurality of first sheet members in a stacking direction and causing the second connecting conductor to penetrate through a plurality of second sheet members in a stacking direction, thereby facilitating electrical connection of the plurality of first electrode layers and electrical connection of the plurality of second electrode layers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a sectional view of a power generation device according to a first embodiment of the present invention;

[0019] FIG. 2 is an exploded perspective view of a first sheet member configuring the power generation device according to the first embodiment of the present invention;

[0020] FIG. 3 is an exploded perspective view of a second sheet member configuring the power generation device according to the first embodiment of the present invention;

[0021] FIG. 4 is an exploded perspective view of the power generation device according to the first embodiment of the present invention;

[0022] FIG. 5 is an exploded perspective view of a power generation device according to a second embodiment of the present invention;

[0023] FIG. 6 is a sectional view of a power generation device according to a third embodiment of the present invention;

[0024] FIG. 7 is a sectional view of a modification of the power generation device according to the third embodiment of the present invention; and

[0025] FIG. 8 is a sectional view of a modification of the power generation device according to the third embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

[0026] FIGS. 1 to 4 illustrate a first embodiment of the present invention. FIG. 1 is a sectional view of a power generation device, FIG. 2 is an exploded perspective view of a first sheet member configuring the power generation device, FIG. 3 is an exploded perspective view of a second sheet member configuring the power generation device, and FIG. 4 is an exploded perspective view of the power generation device.

[0027] As illustrated in FIG. 1, a power generation device 1 according to the present embodiment includes a plurality of first sheet members 10 on a positively charged side, a plurality of second sheet members 20 on a negatively charged side, a first connecting conductor 30 connected to a first electrode layer, which will be described later, of each of the plurality of first sheet members 10, a second connecting conductor 40 connected to a second electrode layer, which will be described later, of each of the plurality of second sheet members 20, a plurality of first insulating members 50 surrounding an outer peripheral portion of the first connecting conductor 30, a plurality of second insulating members 60 surrounding an outer peripheral portion of the second connecting conductor 40, a first conducting wire 70 connected to the first connecting conductor 30, and a second conducting wire 80 connected to the second connecting conductor 40.

[0028] As illustrated in FIG. 2, each of the plurality of first sheet members 10 includes a first base material layer 11, a pair of first electrode layers 12 disposed on both sides of the first base material layer 11 in a thickness direction, and a pair of first chargeable layers 13 disposed on outer sides of the pair of first electrode layers 12 in the thickness direction, respectively.

[0029] The first base material layer 11 is made of a member with an insulating property such as a polyethylene terephthalate, for example.

[0030] The pair of first electrode layers **12** are made of members with conductivity such as conductive non-woven clothes formed by depositing metal such as copper or aluminum on nonwoven clothes, copper foils, or aluminum foils, for example.

[0031] The pair of first chargeable layers **13** are made of members that have an insulating property and are charged in a polarity opposite to that of second chargeable layers, which will be described later. A material of the first chargeable layers **13** is a film-shaped member that is likely to be positively charged, such as polyamide, paper (cellulose), an aluminum foil, or a copper foil, for example. In the present embodiment, the first chargeable layers **13** are preferably made of polyamide.

[0032] As illustrated in FIG. **3**, each of the plurality of second sheet members **20** includes a second base material layer **21**, a pair of second electrode layers **22** disposed on both sides of the second base material layer **21** in the thickness direction, and a pair of second chargeable layers **23** disposed on outer sides of the pair of second electrode layers **22** in the thickness direction, respectively.

[0033] The second base material layer **21** is made of a member with an insulating property similar to that of the first base material layer **11**.

[0034] The pair of second electrode layers **22** are made of members with conductivity similar to that of the first electrode layers **12**.

[0035] The pair of second chargeable layers **23** are made of members that have an insulating property and are charged in a polarity opposite to that of the first chargeable layers **13**. A material of the second chargeable layers **23** is a film-shaped member that is likely to be negatively charged, such as polyimide with a porous structure, polytetrafluoroethylene (PTFE), polystyrene, polyethylene terephthalate, or polyethylene, for example. In the present embodiment, the second chargeable layers **23** are preferably made of polyimide with a porous structure.

[0036] As illustrated in FIG. **1**, the plurality of first sheet members **10** and the plurality of second sheet members **20** are stacked such that parts of them are alternately arranged. In this manner, a state where the second chargeable layers **23** of the second sheet members **20** face the first chargeable layers **13** of the plurality of first sheet members **10**, respectively, is achieved.

[0037] The first connecting conductor **30** is made of a rod-shaped or plate-shaped member with conductivity such as copper or aluminum, for example. As illustrated in FIG. **4**, the first connecting conductor **30** has a shape of a so-called staple used with a stapler including a pair of penetrating portions **31** that penetrate, in a stacking direction, parts of the plurality of first sheet members **10** where the second sheet members **20** are not stacked in a longitudinal direction and a coupling portion **32** that couples proximal end portions of the pair of penetrating portions **31** to each other. The stacked state of the plurality of stacked first sheet members **10** is maintained by causing the pair of penetrating portions **31** to penetrate therethrough in the stacking direction and causing the coupling portion **32** to abut on an outer surface of the first sheet member **10** located on the outermost side in the stacking direction. At this time, the plurality of first electrode layers **12** are connected in parallel to the first connecting conductor **30**.

[0038] The second connecting conductor **40** is made of a member similar to that of the first connecting conductor **30**. The second connecting conductor **40** includes a pair of penetrating portions **41** penetrating, in the stacking direction, through a part of each of the plurality of second sheet members **20** where the first sheet members **10** are not stacked in the longitudinal direction and a coupling portion **42** coupling proximal end portions of the pair of penetrating portions **41** to each other. The stacked state of the plurality of stacked second sheet members **20** is maintained by causing the pair of penetrating portions **41** to penetrate therethrough in the stacking direction and causing the coupling portion **42** to abut on an outer surface of the second sheet member **20** located on the outermost side in the stacking direction. At this time, the plurality of second electrode layers **22** are connected in parallel to the second connecting conductor **40**.

[0039] Each of the plurality of first insulating members **50** is made of a member with an insulating property such as polyethylene terephthalate, for example, and is disposed between the plurality of

first sheet members **10**. The penetrating portions **31** of the first connecting conductor **30** penetrating through the plurality of first sheet members **10** penetrate through the plurality of first insulating members **50** as illustrated in FIGS. **1** and **4**. In other words, each of the plurality of first insulating members **50** surrounds an outer peripheral portion of the first connecting conductor **30** located between the plurality of stacked first sheet members **10**. Here, a ratio $Ti1/Ts2$ between the thickness $Ti1$ of the first insulating members **50** and the thickness $Ts2$ of the second sheet members **20** preferably falls within a range of 0.8 or more and 1.2 or less, and more preferably falls within a range of 0.9 or more and 1.1 or less.

[0040] Each of the plurality of second insulating members **60** is made of a member with an insulating property and is disposed between the plurality of second sheet members **20** similarly to the first insulating members **50**. The penetrating portions **41** of the second connecting conductor **40** penetrating through the plurality of second sheet members **20** penetrate through the plurality of second insulating members **60** as illustrated in FIGS. **1** and **4**. In other words, each of the plurality of second insulating members **60** surrounds an outer peripheral portion of the second connecting conductor **40** located between the plurality of stacked second sheet members **20**. Here, a ratio $Ti2/Ts1$ between the thickness $Ti2$ of the second insulating members **60** and the thickness $Ts1$ of the first sheet members **10** preferably falls within a range of 0.8 or more and 1.2 or less, and more preferably falls within a range of 0.9 or more and 1.1 or less.

[0041] The first conducting wire **70** is made of a wire-shaped member with conductivity such as copper or aluminum, for example. The first conducting wire **70** is electrically connected to the first connecting conductor **30** in a state in which the first conducting wire **70** is sandwiched between an outer surface of the first sheet member **10** located on the outermost side in the stacking direction from among the plurality of stacked first sheet members **10** and the coupling portion **32** of the first connecting conductor **30**.

[0042] The second conducting wire **80** is made of a member similar to that of the first conducting wire **70**. The second conducting wire **80** is electrically connected to the second connecting conductor **40** in a state in which the second conducting wire **80** is sandwiched between an outer surface of the second sheet member **20** located on the outermost side in the stacking direction from among the plurality of stacked second sheet members **20** and the coupling portion **42** of the second connecting conductor **40**.

[0043] The power generation device **1** configured as described above generates power by switching between a first state and a second state: in the first state, the plurality of first sheet members **10** and the plurality of second sheet members **20** alternately stacked do not receive an external force acting in a direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other, whereas in the second state, the plurality of first sheet members **10** and the plurality of second sheet members **20** alternately stacked receive an external force acting in the direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other.

[0044] When the power generation device **1** is in the first state, in response to the force acting in the direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other, the first chargeable layers **13** are positively charged and the second chargeable layers **23** are negatively charged by the first chargeable layers **13** and the second chargeable layers **23** rubbing with each other.

[0045] When the power generation device **1** is in the second state, in response to release of the force acting in the direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other, electricity flows in a direction in which potentials become the same between the first electrode layers **12** and the second electrode layers **22**.

[0046] In this manner, the power generation device **1** of the present embodiment is a power generation device **1** including: the plurality of first sheet members **10**; the plurality of second sheet members **20**; the first connecting conductor **30**; and the second connecting conductor **40**, the first

sheet members **10** including the first chargeable layers **13** on the outer surfaces of the first electrode layers **12**, the second sheet members **20** including the second chargeable layers **23** on the outer surfaces of the second electrode layers **22**, the first chargeable layers **13** and the second chargeable layers **23** facing each other due to parts of the plurality of first sheet members **10** and parts of the plurality of second sheet members **20** being alternately stacked, the power generation device **1** being configured to generate power by way of a change in a mutual contact state between the first chargeable layers **13** and the second chargeable layers **23**, the first connecting conductor **30** penetrating other parts of the plurality of first sheet members **10** and being connected to the first electrode layers **12** of the plurality of first sheet members **10**, and the second connecting conductor **40** penetrating through other parts of the plurality of second sheet members **20** and being connected to the second electrode layers **22** of the plurality of second sheet members **20**.

[0047] In this manner, it is possible to connect the plurality of first electrode layers **12** to the first connecting conductor **30** and to connect the plurality of second electrode layers **22** to the second connecting conductor **40** by causing the first connecting conductor **30** to penetrate through the plurality of first sheet members **10** in a stacking direction and causing the second connecting conductor **40** to penetrate through the plurality of second sheet members **20** in the stacking direction, and to thereby facilitate electrical connection of the plurality of first electrode layers **12** and electrical connection of the plurality of second electrode layers **22**.

[0048] Also, the power generation device **1** preferably includes: the first insulating members **50** surrounding the outer peripheral portion of the first connecting conductor **30** located between the plurality of first sheet members **10**; and the second insulating members **60** surrounding the outer peripheral portion of the second connecting conductor **40** located between the plurality of second sheet members **20**.

[0049] In this manner, it is possible to prevent contact between the first connecting conductor **30** and the second electrode layers **22** of the second sheet members **20**, to prevent contact between the second connecting conductor **40** and the first electrode layers **12** of the first sheet members **10**, and to thereby suppress short-circuiting of an electric circuit in the power generation device **1**.

[0050] Preferably, the ratio $Ti1/Ts2$ between the thickness $Ti1$ of the first insulating members **50** and the thickness $Ts2$ of the second sheet members **20** and the ratio $Ti2/Ts1$ between the thickness $Ti2$ of the second insulating members **60** and the thickness $Ts1$ of the first sheet members **10** each fall within a range of 0.8 or more and 1.2 or less.

[0051] In this manner, it is possible to adjust the intervals of the adjacent first sheet members **10** to the thickness $Ts2$ of the second sheet members **20** with the first insulating members **50**, to adjust the intervals of the adjacent second sheet members **20** to the thickness $Ts1$ of the first sheet members **10** with the second insulating members **60**, and to thereby achieve a uniform thickness dimension over the entire power generation device **1**.

[0052] Also, either the first chargeable layers **13** or the second chargeable layers **23** are preferably made of polyimide with a porous structure while the others are made of polyamide.

[0053] In this manner, it is possible to improve a power generation voltage and to further improve the amount of generated power.

[0054] Also, the first connecting conductor **30** preferably include pairs of penetrating portions **31** penetrating through other parts of the plurality of first sheet members **10** in the stacking direction and the coupling portions **32** extending in the plane direction of the first sheet members **10** and coupling the proximal end portions of the pairs of penetrating portions **31** and the second connecting conductor **40** preferably include pairs of penetrating portions **41** penetrating through other parts of the plurality of second sheet members **20** in the stacking direction and the coupling portions **42** extending in the plane direction of the second sheet members **20** and coupling the proximal end portions of the pairs of penetrating portions **41**.

[0055] In this manner, it is possible to connect the plurality of first electrode layers **12** to the first connecting conductor **30** by causing the penetrating portions **31** of the first connecting conductor

30 to penetrate through the plurality of stacked first sheet members **10**, to connect the plurality of second electrode layers **22** to the second connecting conductor **40** by causing the penetrating portions **41** of the second connecting conductor **40** to penetrate through the plurality of stacked second sheet members **20**, and to thereby facilitate connection between the plurality of first electrode layers **12** and the first connecting conductor **30** and connection between the plurality of second electrode layers **22** and the second connecting conductor **40**.

[0056] Also, the power generation device **1** preferably includes: the first conducting wire **70** connected to the first connecting conductor **30**; and the second conducting wire **80** connected to the second connecting conductor **40**, the first conducting wire **70** is preferably sandwiched between the coupling portion **32** of the first connecting conductor **30** and the first sheet member **10**, and the second conducting wire **80** is preferably sandwiched between the coupling portion **42** of the second connecting conductor **40** and the second sheet member **20**.

[0057] In this manner, it is possible to facilitate connection between the first connecting conductor **30** and the first conducting wire **70** and connection between the second connecting conductor **40** and the second conducting wire **80**.

Second Embodiment

[0058] FIG. **5** illustrates a second embodiment of the present invention and is an exploded perspective view of a power generation device. Note that components similar to those in the above embodiment will be denoted by the same reference signs.

[0059] A first connecting conductor **30** and a second connecting conductor **40** forming part of a power generation device **1** according to the present embodiment are thread-shaped members **33** and **43** with which a plurality of first sheet members **10** stacked along with a plurality of first insulating members **50** and a plurality of second sheet members **20** stacked along with a plurality of second insulating members **60** are sewn together, respectively.

[0060] Each of the thread-shaped members **33** and **43** is made of a strand wire obtained by twisting a metal wire of copper or aluminum, for example, and has conductivity.

[0061] A first conducting wire **70** is electrically connected to the first connecting conductor **30** in a state in which the first conducting wire **70** is sandwiched between an outer surface of the first sheet member **10** located on the outermost side in the stacking direction from among the plurality of stacked first sheet members **10** and the thread-shaped member **33**.

[0062] A second conducting wire **80** is electrically connected to the second connecting conductor **40** in a state in which the second conducting wire **80** is sandwiched between an outer surface of the second sheet member **20** located on the outermost side in the stacking direction from among the plurality of stacked second sheet members **20** and the thread-shaped member **43**.

[0063] In this manner, according to the power generation device **1** of the present embodiment, it is possible to connect the plurality of first electrode layers **12** to the first connecting conductor **30** and to connect the plurality of second electrode layers **22** to the second connecting conductor **40** by causing the first connecting conductor **30** to penetrate through the plurality of first sheet members **10** in the stacking direction and causing the second connecting conductor **40** to penetrate through the plurality of second sheet members **20** in the stacking direction, and to thereby facilitate electrical connection of the plurality of first electrode layers **12** and electrical connection of the plurality of second electrode layers **22**, similarly to the above embodiment.

[0064] Also, the first connecting conductor **30** is preferably conductive thread-shaped members **33** with which the plurality of first sheet members **10** are sewn together, and the second connecting conductor **40** is preferably conductive thread-shaped members **43** with which the plurality of second sheet members **20** are sewn together.

[0065] In this manner, it is possible to connect the plurality of first electrode layers **12** to the first connecting conductor **30** by sewing the plurality of stacked first sheet members **10** together with the thread-shaped member **33**, to connect the plurality of second electrode layers **22** to the second connecting conductor **40** by sewing the plurality of stacked second sheet members **20** together with

the thread-shaped member **43**, and to thereby facilitate connection between the plurality of first electrode layers **12** and the first connecting conductor **30** and connection between the plurality of second electrode layers **22** and the second connecting conductor **40**.

[0066] Also, the power generation device **1** preferably includes: the first conducting wire **70** connected to the first connecting conductor **30**; and the second conducting wire **80** connected to the second connecting conductor **40**, the first conducting wire **70** is preferably sandwiched between the thread-shaped member **33** configuring the first connecting conductor **30** and the first sheet member **10**, and the second conducting wire **80** is preferably sandwiched between the thread-shaped member **43** configuring the second connecting conductor **40** and the second sheet member **20**.

[0067] In this manner, it is possible to facilitate connection between the first connecting conductor **30** and the first conducting wire **70** and connection between the second connecting conductor **40** and the second conducting wire **80**.

Third Embodiment

[0068] FIG. **6** illustrates a third embodiment of the present invention and is a sectional view of a power generation device. Note that components similar to those in the above embodiments will be denoted by the same reference signs.

[0069] The power generation device **1** according to the present embodiment includes first sheet members **10** and second sheet members **20** facing each other in the stacking direction, and spacing retention members **90** that retain spacings between the first and second sheet members.

[0070] Each spacing retention member **90** is a plate-shaped insulating member having a width dimension of, for example, 0.5 mm or greater and 10 mm or less and a length dimension substantially equal to the dimension in the transverse direction of the first and second sheet members **10** and **20**. Each spacing retention member **90** has a thickness dimension of, for example, one-half or less of the thickness dimension of the first insulating member **50** and the thickness dimension of the second insulating member **60**.

[0071] The spacing retention members **90** include spacing retention members **90** each of which is disposed between an end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing one side of the end portion in the stacking direction, and spacing retention members **90** each of which is disposed between an end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing one side of the end portion in the stacking direction. Each spacing retention member **90** is fixed to the one side of the end portion of the first sheet member **10** adjacent to the second insulating member **60** or the one side of the end portion of the second sheet member **20** adjacent to the first insulating member **50**.

[0072] In the first state in which an external force is not acting in a direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other, the spacing retention members **90** of the power generation device **1** having the above described configuration restrict contact between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing the one side of the end portion in the stacking direction and contact between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing the one side of the end portion in the stacking direction.

[0073] Thus, likewise to the above-described embodiments, the power generation device **1** of the present embodiment makes it possible to connect the plurality of first electrode layers **12** to the first connecting conductor **30** and to connect the plurality of second electrode layers **22** to the second connecting conductor **40** by causing the first connecting conductor **30** to penetrate through the plurality of first sheet members **10** in the stacking direction and causing the second connecting conductor **40** to penetrate through the plurality of second sheet members **20** in the stacking direction, thereby facilitating electrical connection of the plurality of first electrode layers **12** and electrical connection of the plurality of second electrode layers **22**, similarly to the above

embodiment.

[0074] Preferably, the power generation device **1** of the present embodiment includes the spacing retention members **90** that retain the spacings between the first and second sheet members **10** and **20**, and the spacing retention members **90** include spacing retention members **90** each of which is disposed between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing one side of the end portion in the stacking direction, and spacing retention members **90** each of which is disposed between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing one side of the end portion in the stacking direction.

[0075] In the first state in which an external force is not acting in the direction in which the first chargeable layers **13** and the second chargeable layers **23** are pushed against each other, the above configuration restricts contact between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing the one side of the end portion in the stacking direction and contact between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing the one side of the end portion in the stacking direction, whereby the first chargeable layers **13** and the second chargeable layers **23** that are in a charged state are less likely to be released from the charged state.

[0076] The third embodiment has been described to have the configuration in which the spacing retention members **90** include spacing retention members **90** each of which is disposed between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing one side of the end portion in the stacking direction, and spacing retention members **90** each of which is disposed between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing one side of the end portion in the stacking direction. However, the present invention is not limited to this configuration. It is sufficient that the spacing retention members **90** are each disposed between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing at least one side of the end portion in the stacking direction and/or between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet member **10** facing at least one side of the end portion in the stacking direction.

[0077] As illustrated in FIG. 7, the power generation device **1** may include the spacing retention members **90** each of which is disposed only between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet member **20** facing one side of the end portion in the stacking direction. The power generation device **1** having this configuration can improve power generation efficiency in comparison with a power generation device that does not include the spacing retention members **90**.

[0078] As illustrated in FIG. 8, the spacing retention members **90** of the power generation device **1** may include spacing retention members **90** disposed between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet members **20** respectively facing the opposite sides of the end portion in the stacking direction, and spacing retention members **90** disposed between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet members **10** respectively facing the opposite sides of the end portion in the stacking direction. The power generation device **1** having this configuration can restrict contact between the end portion of the first sheet member **10** adjacent to the second insulating member **60** and the second sheet members **20** respectively facing the opposite sides of the end portion in the stacking direction, and contact between the end portion of the second sheet member **20** adjacent to the first insulating member **50** and the first sheet members **10** respectively facing the opposite sides of the end portion in the stacking direction, thereby making it possible to achieve further improved power generation efficiency.

[0079] Hereinafter, results of a power generation test performed on each of the power generation

device **1** not including the spacing retention members **90** and the power generation devices **1** including the spacing retention members **90** illustrated in FIGS. **6** to **8** will be described.

[0080] The power generation test was performed using a vertical presser and an oscilloscope.

[0081] The vertical presser includes an air cylinder that is driven by compressed air.

[0082] The air cylinder includes a piston rod the tip end of which is provided with a pressing part that applies a compressive force in the thickness direction of the power generation device **1**. The pressing part has a square flat surface having each side of 30 mm.

[0083] The air cylinder fixed to the main body of the vertical presser was in a posture in which its axial direction was along the vertical direction and the pressing part was oriented downward. In a state in which the power generation device **1** was disposed below the pressing part, the air cylinder of the vertical presser is driven, so that the pressing part applied a load of 16 kg to the power generation device **1**, and the application of the load was released 0.3 seconds after the start of the application of the load.

[0084] In the power generation test, a voltage at the start of the application of the load to the power generation device **1** by the vertical presser and a voltage at the release of the application of the load were measured by the oscilloscope, and the power generation output per one second was calculated by the following equation.

[00001]
$$P = \frac{\int_0^1 V(t)^2 dt}{R}$$
 [Formula1]

[0085] In the above equation, P is a power generation output (W), V is a measured voltage (V), R is a probe resistance (Ω) of the oscilloscope, and t is a measurement time at the time of power generation.

[0086] The results of the power generation test were as follows. The power generation output of the power generation device **1** without the spacing retention members **90** was 174 μ W. On the other hand, the power generation output of the power generation device **1** including the spacing retention members **90** arranged as illustrated in FIG. **6** was 699 μ W. The power generation output of the power generation device **1** including the spacing member **90** arranged as illustrated in FIG. **7** was 385 μ W. The power generation output of the power generation device **1** including the spacing retention members **90** arranged as illustrated in FIG. **8** was 731 μ W. That is, the power generation test demonstrated that the power generation output of each of the power generation devices **1** including the spacing retention members **90** illustrated in FIGS. **6** to **8** was higher than that of the power generation device **1** not including the spacing retention member **90**.

EXPLANATION OF REFERENCE NUMERALS

[0087] **1** Power generation device [0088] **10** First sheet member [0089] **12** First electrode layer [0090] **13** First chargeable layer [0091] **20** Second sheet member [0092] **22** Second electrode layer [0093] **23** Second chargeable layer [0094] **30** First connecting conductor [0095] **31** Penetrating portion [0096] **32** Coupling portion [0097] **33** Thread-shaped member [0098] **40** Second connecting conductor [0099] **41** Penetrating portion [0100] **42** Coupling portion [0101] **43** Thread-shaped member [0102] **50** First insulating member [0103] **60** Second insulating member [0104] **70** First conducting wire [0105] **80** Second conducting wire [0106] **90** Spacing retention member

Claims

1. A power generation device comprising: a plurality of first sheet members; a plurality of second sheet members; a first connecting conductor; and a second connecting conductor, each of the first sheet members comprising a first electrode layer and a first chargeable layer disposed on an outer surface of the first electrode layer, each of the second sheet members comprising a second electrode layer and a second chargeable layer disposed on an outer surface of the second electrode layer, the first chargeable layer and the second chargeable layer facing each other due to parts of the plurality of first sheet members and parts of the plurality of second sheet members being

alternately stacked, the power generation device being configured to generate power by way of a change in a mutual contact state between the first chargeable layer and the second chargeable layer, the first connecting conductor penetrating through another part of each of the plurality of first sheet members and being connected to the first electrode layer of each of the plurality of first sheet members, the second connecting conductor penetrating through another part of each of the plurality of second sheet members and being connected to the second electrode layer of each of the plurality of second sheet members.

2. The power generation device according to claim 1, comprising: a first insulating member surrounding an outer peripheral portion of the first connecting conductor, the outer peripheral portion being located between the plurality of first sheet members; and a second insulating member surrounding an outer peripheral portion of the second connecting conductor, the outer peripheral portion being located between the plurality of second sheet members.

3. The power generation device according to claim 2, wherein a ratio $Ti1/Ts2$ between a thickness $Ti1$ of the first insulating member and a thickness $Ts2$ of the second sheet members and a ratio $Ti2/Ts1$ between a thickness $Ti2$ of the second insulating member and a thickness $Ts1$ of the first sheet members are each 0.8 or more and 1.2 or less.

4. The power generation device according to claim 1, wherein one of the first chargeable layer and the second chargeable layer is made of polyimide with a porous structure while the other is made of polyamide.

5. The power generation device according to claim 1, wherein the first connecting conductor comprises a pair of penetrating portions penetrating through other parts of the plurality of first sheet members in a stacking direction, and a coupling portion extending in a plane direction of the first sheet members and coupling proximal end portions of the pair of penetrating portions to each other, and the second connecting conductor comprises a pair of penetrating portions penetrating through other parts of the plurality of second sheet members in a stacking direction, and a coupling portion extending in a plane direction of the second sheet members and coupling proximal end portions of the pair of penetrating portions to each other.

6. The power generation device according to claim 5, comprising: a first conducting wire connected to the first connecting conductor; and a second conducting wire connected to the second connecting conductor, wherein the first conducting wire is sandwiched between the coupling portion of the first connecting conductor and the first sheet member, and the second conducting wire is sandwiched between the coupling portion of the second connecting conductor and the second sheet member.

7. The power generation device according to claim 1, wherein the first connecting conductor is a conductive thread-shaped member with which the plurality of first sheet members are sewn together, and the second connecting conductor is a conductive thread-shaped member with which the plurality of second sheet members are sewn together.

8. The power generation device according to claim 7, comprising: a first conducting wire connected to the first connecting conductor; and a second conducting wire connected to the second connecting conductor, wherein the first conducting wire is sandwiched between a thread-shaped member configuring the first connecting conductor and the first sheet member, and the second conducting wire is sandwiched between a thread-shaped member configuring the second connecting conductor and the second sheet member.

9. The power generation device according to claim 2, comprising: spacing retention members that retain spacings between the first and second sheet members facing each other in a stacking direction, wherein the spacing retention members are each disposed between an end portion of the first sheet member adjacent to the second insulating member and the second sheet member facing at least one side of the end portion in the stacking direction and/or between an end portion of the second sheet member adjacent to the first insulating member and the first sheet member facing at least one side of the end portion in the stacking direction.

